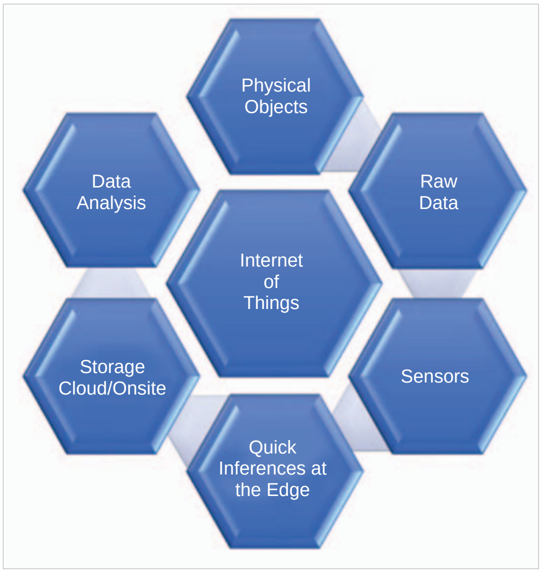

Figure 1: A typical Internet of Things setup

platform. Once you have entered your profile settings, you can move on to the statistics, as shown in Figure 2.

Go to the *Devices* tab and create a device. This will ask for more details, as shown in Figure 3.

The device type is generally a generic Thinger device (Wi-Fi, Ethernet, GSM). However, the user has the option to select HTTP devices (Sigfox, Lora, cURL)/ MQTT device/ NB-IoT. The Device ID is important and you have to remember it while interfacing with your code. Device information is optional and is limited to the user (for identification). But Device Credentials are vital. Each device has its own identifier/credential, so that a compromised device does not affect other devices. All your passwords in the server are stored securely using PBKDF2 SHA256 with a 32-byte salt generated with PRNG and a non-depreciable number of iterations. (Please remember, the password cannot be recalled later.)

If the steps have been implemented correctly till now, you should see your device disconnected when you press the *Add Device* button.

Using your new Device ID and the Device Credentials to connect the new device, you will need to install the required libraries or development environment. For the present example, we will be using the Arduino IDE along with an ESP8266 device.

The ESP8266 chip (Figure 4) from Espressif is the new generation of low-cost Wi-Fi chips after the TI CC3000/ CC3200. This small chip not only integrates the entire Wi-Fi features, but also a powerful programmable MCU. Depending on the board layout (ESP-01, ESP-03, ESP-07, ESP12, etc), it is attached to a programmable flash, ranging from 512K to 4M. This increases the available user code space, and makes possible other cool features like a small file system or OTA updates. I am using the ESP 12 module for the present illustration.

The device can be directly programmed from the Arduino IDE. The very first requirement is to install the board via the Arduino Boards Manager (Figure 5). From the *File* menu, go to *Preferences*, and enter *http://arduino.esp8266.com/stable/package_esp8266com_index.json* in the Additional Boards Manager URL.

The next step involves installing the ESP8266 board. Go to *Tools* > *Boards*

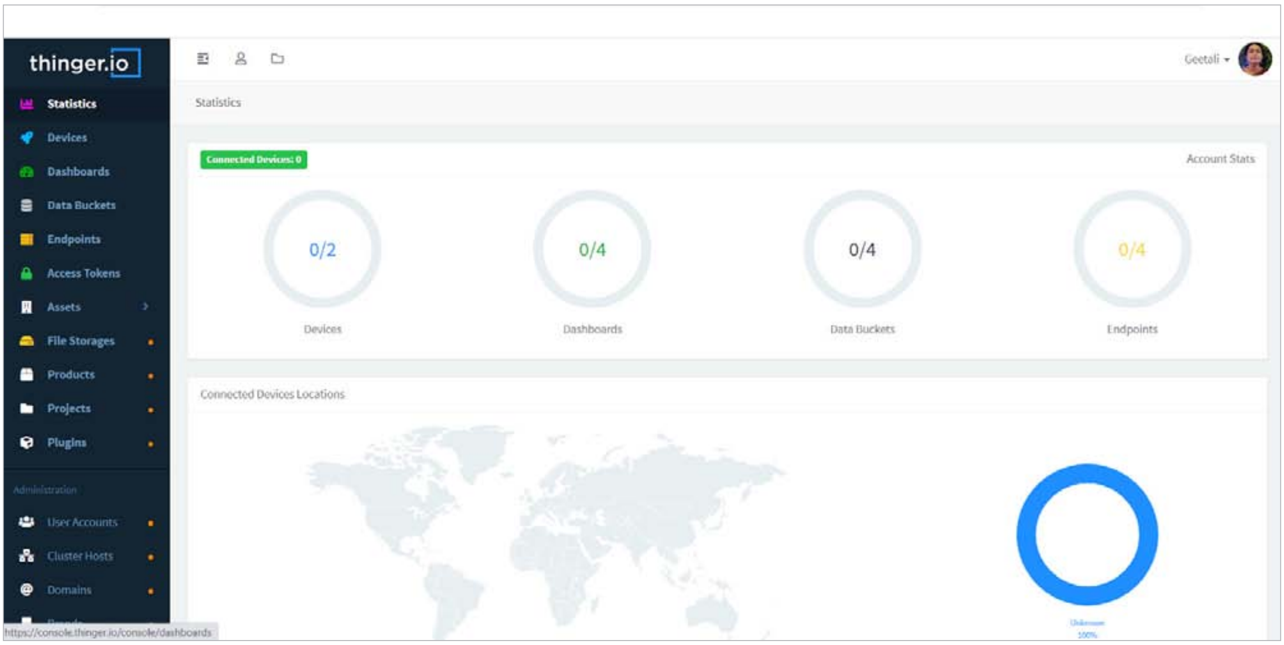

Figure 2: View of the statistics